Lecture 19: Sections 7.1 and 7.2

Section 7.1 Linear and Nonlinear Systems of Equations

Linear systems - solve by graphing

Linear systems - solve by substitution method

Nonlinear system of equations

Applications

Section 7.2 Two Variable Linear Systems

Linear systems - solve by elimination method

Linear systems - solve by graphing

Applications

Tough [illegible] Fri 4th CAR

Lecture 19: Sections 7.1 and 7.2
System of Equations

Def. A **system of equations** is a collection of two or more equations, each containing one or more variables.

A **solution** of a system of equations consists of values of the variables that satisfy ~~each~~ both **equations** in the system. To **Solve** a system is to find all solutions of the system.

A system is **consistent** if it has at least one solution; otherwise it is called **inconsistent**.

Def. A system of the form

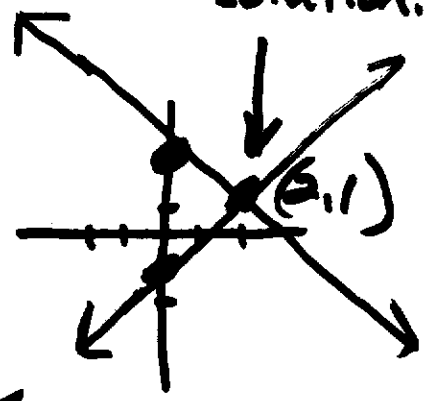
x and y are to the first power.

$$\begin{cases} a_1x + b_1y = c_1 \\ a_2x + b_2y = c_2 \end{cases}$$

is a **system of two linear equations in two variables**.

The solution set of the system is all ordered pairs (x, y) that satisfy the system.

common solution.

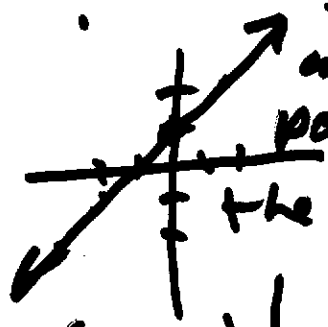


Solving by Graphing

ex. Solve each system: solve for y.

1) $\begin{cases} x + 2y = 4 & y = -\frac{1}{2}x + 2 \\ x - y = 1 & y = x - 1 \end{cases}$

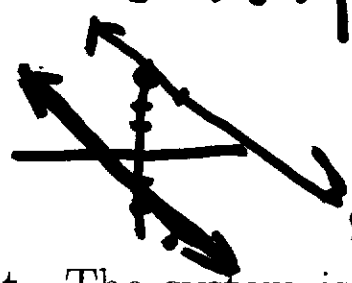
2) $\begin{cases} x - y = -1 & y = x + 1 \\ -2x + 2y = 2 & y = x + 1 \end{cases}$



solutions are any points on the line.

$\{(x,y) \mid x - y = -1\}$

3) $\begin{cases} x + y = 3 & y = -x + 3 \\ x + y = -2 & y = -x - 2 \end{cases}$



No solution since lines are parallel!

Note: same slope, but different y-intercepts

We have three possibilities:

1. The lines intersect at one point. The system is consistent with exactly one solution. The equations are **independent**.

2. The lines coincide. There are infinitely many solutions (every point one the line); the system is consistent and the equations are **dependent**.

3. The lines are parallel. The system is inconsistent, having no solutions.

The Method of Substitution

1. **Solve** for one variable in an equation in terms of the other.

2. **Substitute** the result in the other equation.

3. Solve the resulting equation in one variable.

4. **Back-substitute** into the equation from (1) to find the value of the second variable.

5. Check your solution in **each** equation.

<u>ex.</u> Solve the system $\begin{cases} x - 2y = 1 \\ 4x - 3y = -1 \end{cases}$ $x = 2y + 1$

$4(2y+1) - 3y = -1$

$8y + 4 - 3y = -1$

$5y = -5$

$y = -1$

$x = 2y + 1$

$x = 2(-1) + 1 = -1$

Easy to check.

thus $(-1, -1)$ is the common solution.

The Method of Elimination

① Vertically line up the xs, ys, =s, and the constants.

Rewrite the equations as necessary so that we can add or subtract two equations to eliminate a variable.

② Get opposite coef. on one of the variables

ex. Solve the system:

③ Add the equations together.

1) $\begin{cases} 3x + 2y = 4 \\ 5x - 2y = 8 \end{cases}$

$8x = 12$

$x = \frac{12}{8} = \frac{3}{2}$

④ Substitute back into one of the original equations to find the other variable.

$3\left(\frac{3}{2}\right) + 2(y) = 4$

$2y = 4 - \frac{9}{2}$

$\frac{1}{2}[2y = -\frac{1}{2}]$

$y = -\frac{1}{4}$

thus $\boxed{\left(\frac{3}{2}, -\frac{1}{4}\right)}$

easy to check.

2) $\begin{cases} 6\left(\frac{x}{2} - \frac{2y}{3} = -2\right) \\ -6x + 8y = 24 \end{cases}$

$3x - 4y = -12$

$\frac{1}{2}[-6x + 8y = 24]$

$$\begin{array}{r} 3x - 4y = -12 \\ -3x + 4y = 12 \\ \hline 0 = 0 \end{array}$$

Since we lost the variables and the resulting equation is true, the equations represent the same line.

Thus $\{(x, y) \mid -6x + 8y = 24\}$

Nonlinear System of Equations

A **nonlinear system** is a system of equations in which at least one equation is not linear.

<u>**ex.**</u> Solve each of the following systems. Graph each equation and verify the points of intersection.

1) $\begin{cases} y = x^2 + 1 \\ y = x + 1 \end{cases}$

$x^2 + 1 = x + 1$

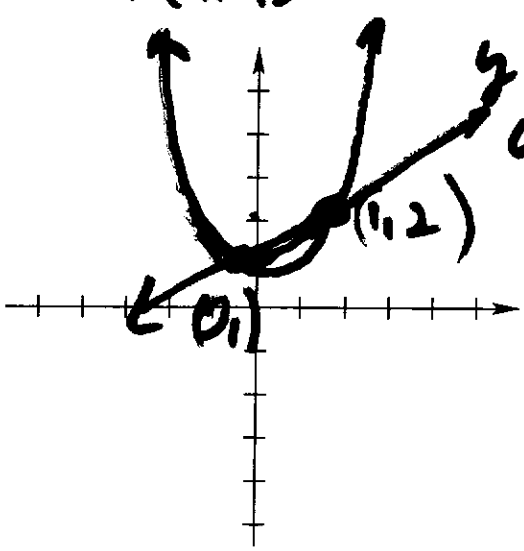
$x^2 - x = 0$

$x(x-1) = 0$ $\quad x = 0 \quad x = 1$

$y = 0 + 1 \quad y = 1 + 1$

$y = 1 \quad y = 2$

$\boxed{(0,1)} \quad \boxed{(1,2)}$



2) $\begin{cases} x^2 + y^2 = 25 \\ -2x + y = -5 \end{cases}$ $y = 2x - 5$

$x^2 + (2x-5)^2 = 25$

$x^2 + 4x^2 - 20x + \cancel{25} = \cancel{25}$

$x^2 + 4x^2 - 20x = 0$

$5x^2 - 20x = 0$

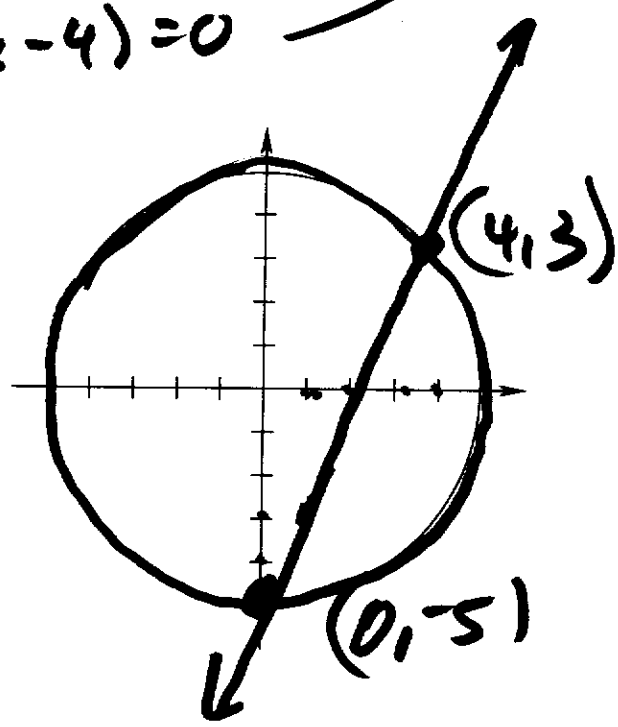
$5x(x-4) = 0$

$x = 0$ $\quad$ $x = 4$

$-2(0) + y = -5$ $\quad$ $-2(4) + y = -5$

$y = -5$ $\quad$ $y = 3$

$(0, -5)$ $\quad$ $(4, 3)$



$x = 3$ one sol. $\quad x = 3$ one sol.

$x^2 = 9$ two sol. $\quad x^3 = 27$ one sol. $\rightarrow \boxed{(1,1)}$

Must check solutions when you raise both sides to an even power.

$1 = \sqrt{1}$ ✓

$1 = 2 - 1$ ✓

$\cancel{(4,-2)}$

$\cancel{-2 \neq \sqrt{4}}$

3) $\begin{cases} y = \sqrt{x} \\ y = 2 - x \end{cases}$ $\quad y = -x + 2$

$(\sqrt{x})^2 = (2-x)^2$

$x = 4 - 4x + x^2$

$x^2 - 5x + 4 = 0$

$(x-1)(x-4) = 0$

$x = 1 \quad x = 4$

$y = 1 \quad y = -2$

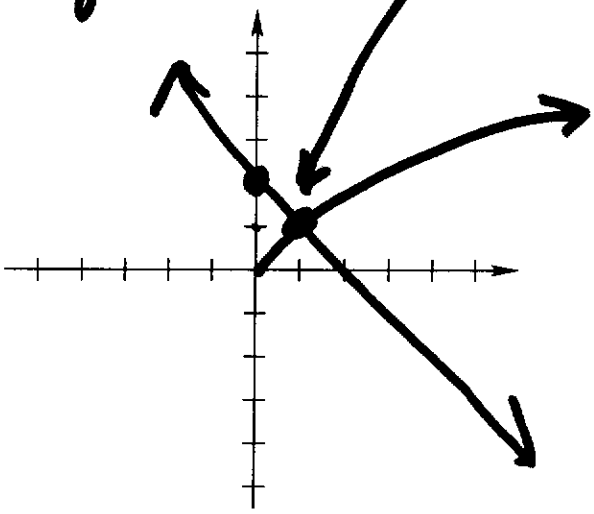
$\cancel{(4,2)}$

$2 = \sqrt{4}$ ✓

$\cancel{2 = -2 - 4}$

$2 \neq -2$

common sol. $(1,1)$



Applications

ex. Steve made a profit of \$25,000 from the sale of his condominium. He then invested part of that money in a bond fund at 4% interest and the rest in a mutual fund which returns 6% interest. After a year, he had earned \$1260. How much did he invest in each account?

Let x = amt in the 4% account.
y = amt in the 6% account.

total money $x + y = \$25{,}000$

amts. invested $[.04x + .06y = \$1260]\,100$

$$4x + 6y = 126{,}000$$
$$x + y = 25{,}000$$

Solve to get $\boxed{\begin{array}{l} x = \$12{,}000 \\ y = \$13{,}000 \end{array}}$

$3\text{hr } 36\text{min} = 3\frac{36}{60} = 3\frac{6}{10} = 3.6$ [redacted] hrs.

<u>ex.</u> An airplane flying into a headwind makes the 1800 miles trip between Phoenix and Pittsburg in 3 hours and 36 minutes. the return trip takes 3 hours. Find the speed of the plane and the speed of the wind, assuming that both remain constant.

$x =$ speed of plane
$y =$ speed of wind

[redacted]

$$d = r \cdot t$$

going into headwind: $\left[1800 = (x-y) \cdot (3.6)\right] \frac{1}{3.6}$

with the wind : $\left[1800 = (x+y) \cdot (3)\right] \frac{1}{3}$

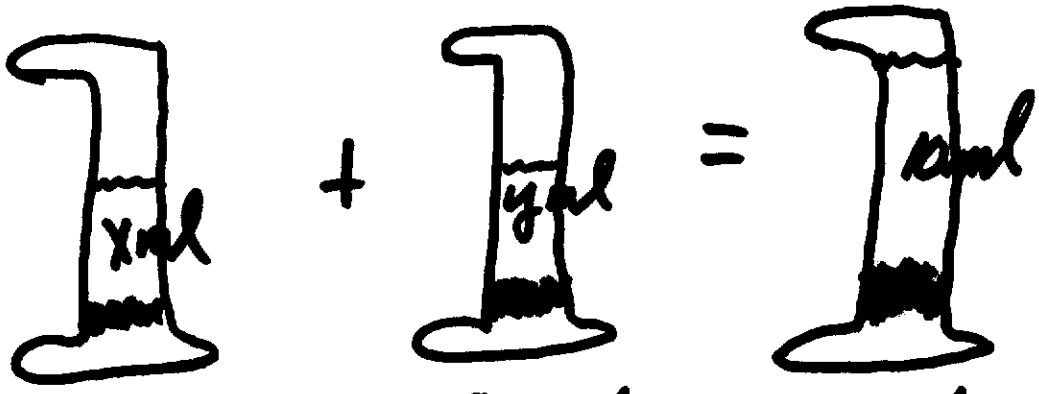
$$500 = x - y$$
$$600 = x + y$$
$$1100 = 2x$$

[redacted]

$$\boxed{\begin{aligned} x &= 550 \text{ mph} \\ y &= 50 \text{ mph} \end{aligned}}$$

ex. A chemist has one solution containing a 10% concentration of acid and a second solution containing a 15% concentration of acid. How many milliliters (ml) of each should be mixed in order to obtain 10 ml of a solution containing a 12% acid concentration?

Let x = ml of 10% acid sol.
y = ml of 15% acid sol.
want 10ml of 12% acid sol.



10% acid sol 15% acid sol 12% acid sol

$x + y = 10$ volume

Solve to get $x = 6$, $y = 4$

$10\%(x) + 15\%(y) = 12\%(10)$ pure acid

$100[.1x + .15y = .12(10)] \rightarrow 10x + 15y = 120$

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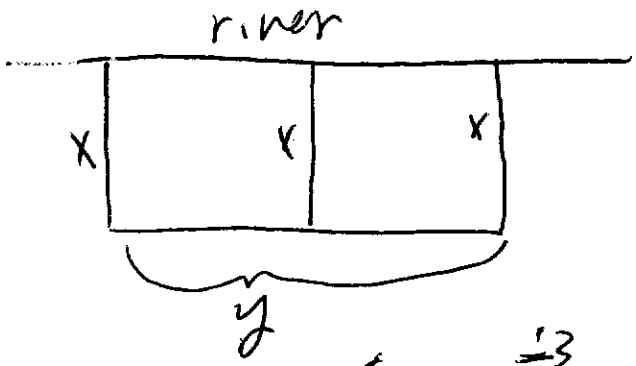
ex. A farmer has \$1500 available to build a fence along a straight river to create two identical rectangular pastures. The material for side parallel to the river costs \$6 per foot, and materials for the three sides perpendicular to the river costs \$5 per foot. Find the dimensions of the pasture if no fence is needed along the river and the total area of the pastures is 6250 ft^2.



cost: $\$5(3x) + \$6(y) = \$1500 \xrightarrow{\div 3}$ $5x + 2y = 500$

Solve for y, then sub into area equation.

area: $x \cdot y = 6250$

$2y = -5x + 500$

$y = -\frac{5}{2}x + 250$

so $x\left(-\frac{5}{2}x + 250\right) = 6250$

$-\frac{2}{5}\left[-\frac{5}{2}x^2 + 250x - 6250 = 0\right]$

$x^2 - 100x + 2500 = 0$

$(x-50)^2 = 0$

$\boxed{x = 50}$

$y = -\frac{5}{2}(50) + 250 = 125$

$(y = 125)$

Practice.

Solve the system:

1) $\begin{cases} 0.4x - 0.2y = 1 \\ 2x = y + 3 \end{cases}$

2) $\begin{cases} 5x - 3y = 16 \\ 2x - 4y = 12 \end{cases}$

Answer. 1) $\emptyset$ 2) $(2, -2)$